

07. Vaginitis: The Vaginal Ecosystem and Diagnostic Discipline

Companion reading handout for simultaneous listening.

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The first mistake in vaginitis is believing the discharge.

Discharge matters. Odor matters. Itching matters. Burning matters. But none of these, by themselves, is a diagnosis. They are the beginning of the evaluation, not the end of it.

This is why vaginitis is such a good test of clinical discipline. The patient may come in saying, "I have yeast." The chart may say "recurrent vaginitis." The discharge may look thick, gray, watery, yellow, or almost normal. But if the resident lets the first label become the final diagnosis, the visit can go wrong quickly.

The patient with bacterial vaginosis may be treated again and again for yeast. The patient with trichomoniasis may have a negative wet mount and be falsely reassured. The postmenopausal patient with irritation and a high pH may be treated for bacterial vaginosis when the real problem is estrogen deficiency or vulvar disease. The patient with contact dermatitis may receive repeated antifungals, and each new medication may worsen the irritation.

So the dominant model for this lecture is simple.

Diagnose vaginitis by four layers: location, ecosystem, organism, and inflammation.

Location asks: is this mainly vulvar, vaginal, anal, urinary, cervical, or pelvic?

Ecosystem asks: is the normal lactobacillus-dominant acidic environment intact?

Organism asks: is there evidence of bacterial vaginosis, *Trichomonas vaginalis*, or *Candida*?

Inflammation asks: are the tissues reacting, or is this mostly a microbiome shift without much tissue inflammation?

If you keep those four layers in mind, vaginitis stops being a vague discharge complaint and becomes a structured bedside diagnosis.

This episode is about diagnosis. Treatment, recurrence, resistance, and partner management come in the next episode.

The normal reproductive-age vagina

Estrogen increases glycogen inside vaginal epithelial cells. Glycogen supports lactobacilli. Lactobacilli produce lactic acid. Lactic acid keeps the vaginal pH less than four point five.

That acidic environment is not just a number on pH paper. It is the chemical signature of a working ecosystem. It tells you that estrogenized epithelium is present, lactobacilli are active, and the environment is suppressing many organisms that would otherwise overgrow.

But normal vaginal flora are not sterile and not uniform. *Gardnerella vaginalis*, *Escherichia coli*, group B streptococci, genital *Mycoplasma* species, and *Candida albicans* can all be found in the vagina. So the question is not simply, "Can I detect an organism?" The better question is, "Does this organism, in this patient, with this ecosystem and this inflammation pattern, explain the symptoms?"

That distinction prevents a large amount of overtreatment.

Now contrast the estrogenized vagina with the prepubertal or postmenopausal vagina.

When estrogen is low, the epithelium is thinner. Glycogen is lower. Lactobacilli are sparse. Lactic acid production is reduced. pH often rises above four point five.

That high pH does not automatically mean bacterial vaginosis. It may simply mean that the estrogen-lactobacillus-lactic acid system is not present in the same way. ACOG emphasizes that bacterial vaginosis and vaginal candidiasis are less common in prepubertal girls and in postmenopausal women who are not using estrogen.

That is the first major resident trap.

pH tells you the ecosystem. It does not name the organism.

So if a postmenopausal patient has burning, irritation, dyspareunia, thin tissues, and high pH, do not rush to call it bacterial vaginosis. Think about genitourinary syndrome of menopause. Think about vulvar dermatoses. Think about irritants. Think about whether the exam and microscopy actually support infection.

Move to the initial evaluation

ACOG recommends a complete history, physical examination of the vulva and vagina, and clinical testing of vaginal discharge. The core office tests are pH testing, the potassium hydroxide whiff test, and microscopy with saline and ten percent KOH.

Commercial tests can be useful, especially when microscopy is not available. Molecular panels can be helpful in selected settings. But the resident should not lose

the bedside model. The model is still location, ecosystem, organism, and inflammation.

History begins by localizing the symptom.

Ask where the problem is. Is it external vulvar itch? Deep vaginal discharge? Burning at the introitus? Anal irritation? Urinary dysuria? Pelvic pain? Postcoital bleeding? A patient may use one word, like "vaginitis," to describe several different anatomic problems.

Then ask about duration, recurrence, relation to menses, sexual history, new partners, prior sexually transmitted infections, self-treatment, over-the-counter antifungals, douching, soaps, perfumes, spermicides, lubricants, pads, wipes, and topical products.

Ask about diabetes, HIV, immunosuppressive therapy, inflammatory bowel disease, and recent antibiotics. These are not decorative history questions. They change your probability map. Diabetes and immunosuppression matter for candidiasis. Sexual exposure matters for trichomoniasis. Douching and intravaginal products matter for bacterial vaginosis and irritant disease.

And self-treatment matters because repeated treatment can obscure the original presentation. A patient who has used multiple antifungals may no longer have a clean untreated clinical picture. She may also have irritation from the medications themselves.

The resident reflex is this: before naming the infection, localize the symptom.

Examine before you label

Start with the vulva and anus, not only the speculum. Look for erythema, edema, fissures, excoriations, hypopigmentation, plaques, papules, erosions, architectural change, and perianal involvement.

This is where bacterial vaginosis, candidiasis, and trichomoniasis begin to separate.

Bacterial vaginosis is usually not a vulvar inflammatory disease. It is a vaginal dysbiosis. It may cause discharge and odor, but it should not typically produce marked vulvar edema, fissures, or intense external inflammation.

Candidiasis can inflame the vulva. Severe candidiasis can produce erythema, edema, excoriations, and fissures.

Trichomoniasis can inflame the vagina and cervix. There may be erythema, petechiae, discharge, odor, pruritus, irritation, dysuria, or postcoital bleeding.

So the vulvar exam is not optional. It is part of the diagnostic logic.

During the speculum exam, discharge appearance can give clues, but it is not diagnostic alone. Thin gray homogeneous discharge and fishy odor suggest bacterial vaginosis. Yellow-green frothy discharge suggests trichomoniasis. Thick white discharge can occur with candidiasis. But these are patterns, not proof.

The next step is to test the environment.

Take the pH sample from the mid-vaginal sidewall. This detail matters. Cervical mucus, blood, semen, lubricants, and other substances can falsely elevate pH. If you sample the wrong place, you may create a false diagnostic trail.

Then perform the whiff test with ten percent KOH. An amine or fishy odor supports bacterial vaginosis, though it is not the whole diagnosis.

Then microscopy.

With zero point nine percent saline, look for clue cells, motile trichomonads, inflammatory cells, lactobacilli, and the general cellular background. With ten percent KOH, look for yeast forms: budding yeast, spores, pseudohyphae, or hyphae. The KOH helps dissolve background cellular material and makes fungal elements easier to see.

Put the pieces together

Normal physiologic discharge

Normal discharge may be white, creamy, or clear. pH is usually three point five to four point five. Microscopy shows mature squamous cells, lactobacilli, and few inflammatory cells. The key is that there is no pathogen-driven pattern, no prominent inflammation, and no dysbiosis pattern.

The resident should be comfortable telling a patient when discharge is physiologic. Not every discharge complaint needs antimicrobial treatment. Sometimes the best care is explanation, reassurance, and avoidance of unnecessary therapy.

Bacterial vaginosis

Bacterial vaginosis is the classic example of why the word "infection" can mislead. It is not a classic invasive inflammatory infection. It is a microbiome shift.

Lactobacilli decrease. Facultative anaerobes and other bacterial vaginosis-associated organisms increase. The acidic environment is disrupted. pH rises above four point five. Amine odor may appear. Squamous cells become covered with adherent coccobacilli, creating clue cells.

The patient may report thin white-gray homogeneous discharge. She may notice fishy odor, especially after intercourse or around menses. But many patients with bacterial

vaginosis are asymptomatic.

That matters because bacterial vaginosis is often overdiagnosed. Clinical correlation is necessary. The goal is not to treat every microbiologic possibility. The goal is to diagnose the syndrome that explains the patient's presentation.

ACOG recommends diagnosing bacterial vaginosis with Amsel criteria or Gram stain with Nugent scoring.

Amsel criteria require three of four findings.

One, homogeneous thin white-gray discharge.

Two, more than twenty percent clue cells on saline microscopy.

Three, vaginal pH greater than four point five.

Four, a positive KOH whiff test.

Clue cells are especially useful. They are vaginal squamous epithelial cells studded with adherent bacteria. They are not just "bacteria seen on microscopy." They are epithelial cells whose borders are obscured by adherent organisms, which reflects the dysbiosis pattern.

Amsel criteria have reported sensitivity around ninety two percent and specificity around seventy seven percent compared with Gram stain with Nugent scoring. Nugent scoring assigns points based on bacterial morphotypes on Gram stain. Zero to three is normal flora. Four to six is intermediate flora. Seven to ten is bacterial vaginosis flora.

Nugent is the reference framework, but it is often not practical for routine office care. Amsel is the clinical workhorse.

The point that residents must not miss

Routine bacterial culture is not recommended for the diagnosis of bacterial vaginosis. The normal vaginal flora are heterogeneous. Detecting *Gardnerella* or other organisms does not prove bacterial vaginosis by itself. Bacterial vaginosis is a dysbiosis pattern, not a culture result.

The resident reflex is this: BV is high pH, clue cells, whiff, and dysbiosis. It is not diagnosed by culture.

Trichomoniasis

Trichomoniasis is different. It is caused by *Trichomonas vaginalis*, and it is a sexually transmitted infection. It may cause discharge, odor, pruritus, irritation, dysuria, postcoital bleeding, and vaginal or cervical erythema. Classically, discharge may be

yellow-green and frothy, and the cervix may have petechiae. But more than half of patients may be asymptomatic or minimally symptomatic.

That combination is clinically important. Trichomoniasis can be present without dramatic symptoms, and a diagnosis does not necessarily prove recent acquisition. Asymptomatic carriage can persist.

On testing, trichomoniasis usually has pH greater than four point five. Microscopy may show motile trichomonads and abundant inflammatory cells. But wet mount sensitivity is limited, only about fifty to sixty percent.

That is one of the highest-yield traps in the whole lecture.

A negative wet mount does not rule out trichomoniasis.

Nucleic acid amplification testing is recommended for diagnosis. NAAT can be performed on vaginal, cervical, or urine specimens, and ACOG reports sensitivity and specificity in the range of roughly ninety five to one hundred percent. That is why molecular testing changed the diagnostic standard. It is not because microscopy is useless. It is because microscopy misses too much disease.

Contrast trichomoniasis with bacterial vaginosis

Both can have elevated pH. Both can have odor. Both can have discharge. But they are not the same clinical entity.

Bacterial vaginosis is a dysbiosis pattern associated with sexual activity, but it is not caused by one sexually transmitted pathogen. Trichomoniasis is a sexually transmitted infection. That difference changes counseling, partner management, retesting, and public health implications, which we will build in the treatment episode.

For diagnosis, the reflex is this: if trichomoniasis is plausible, do not let a negative wet mount end the evaluation. Use NAAT.

Candidiasis

Vulvovaginal candidiasis is inflammation and infection with *Candida* species. It is common; many women report at least one lifetime episode. But it is also overdiagnosed, especially when itching is treated as if it automatically means yeast.

Symptoms may include pruritus, burning, dysuria, dyspareunia, vulvar edema, erythema, fissures, excoriations, and abnormal discharge. The discharge may be thick and white, but it may also look normal.

Here is the diagnostic rule.

Symptoms alone are not enough.

In a symptomatic patient, diagnosis requires one of two things: visualization of spores, pseudohyphae, or hyphae on wet mount microscopy, or a positive vaginal fungal culture or commercial diagnostic test for *Candida* species.

pH is usually normal, three point five to four point five. That is a useful contrast. Bacterial vaginosis and trichomoniasis usually push pH above four point five. Candidiasis often causes marked vulvar symptoms while the acidic lactobacillus-dominant environment remains present.

So *Candida* teaches a very important idea.

The intensity of symptoms does not always track with pH.

A patient can have severe vulvar burning, edema, and fissures with a normal pH. That does not make the symptoms mild. It tells you that the problem is not primarily loss of the acidic vaginal ecosystem.

Microscopy is useful, but it is not perfect. Yeast microscopy sensitivity is only about fifty to seventy percent. If symptoms persist and microscopy is negative, culture is preferred. Culture is also important in recurrent or resistant disease because species matters.

Candida albicans is the usual organism and is generally susceptible to standard azole therapy. But non-*albicans* *Candida*, especially *Candida glabrata*, may be harder to recognize on microscopy and less responsive to usual azoles. *Candida glabrata* may not form pseudohyphae, so the microscopy can be less obvious.

Be careful with culture

Culture can detect yeast in symptomatic patients when microscopy is negative. But cultures may be positive in up to about thirty percent of asymptomatic patients. So a positive culture does not replace clinical judgment. It must fit the symptoms and exam.

The resident reflex is this: candidiasis is symptoms plus objective yeast, usually with normal pH. It is not itching alone.

Let us rehearse the model through cases

First case.

A reproductive-age patient has thin gray discharge and fishy odor after intercourse. The vulva is not markedly inflamed. pH is above four point five. The whiff test is positive. Saline microscopy shows clue cells in more than twenty percent of epithelial cells and few inflammatory cells.

This is bacterial vaginosis. The diagnosis rests on the dysbiosis pattern: elevated pH, whiff, clue cells, and compatible discharge. You do not need a bacterial culture. In fact, culture would be the wrong diagnostic instinct.

Second case.

A patient has yellow-green discharge, irritation, and dysuria. pH is above four point five. Microscopy shows many inflammatory cells, but no motile trichomonads are seen.

Do not stop there. The wet mount missed possibility is real. Because wet mount sensitivity is limited, the next diagnostic move is NAAT for *Trichomonas vaginalis*. This is exactly the kind of patient where "I did not see it moving" is not strong enough.

Third case.

A patient has intense vulvar itching, external burning, fissures, and dyspareunia. pH is four point zero. KOH microscopy does not show hyphae.

This is not automatically "no yeast." It is also not automatically "treat yeast anyway forever." Persistent symptoms with negative microscopy should lead to culture or an appropriate commercial test, plus a careful search for mimics: contact dermatitis, vulvar dermatosis, irritant exposure, and other causes of vulvar pain or inflammation.

Fourth case.

A postmenopausal patient not using estrogen has burning and dyspareunia. pH is elevated. The epithelium looks thin. Lactobacilli are sparse.

The elevated pH is real, but it does not automatically diagnose bacterial vaginosis. The estrogen-depleted vagina normally loses the lactobacillus-lactic acid pattern. Think about genitourinary syndrome of menopause and vulvar disease before prescribing antibiotics.

These cases all point to the same lesson.

pH is powerful, but only when interpreted in context.

Now a few diagnostic traps deserve explicit attention.

Trap one: self-diagnosis.

Patients are often experienced with their own symptoms, and their history matters. But self-diagnosis of vaginitis is not reliable enough to replace evaluation, especially when symptoms recur, fail therapy, or differ from prior episodes. Repeated over-the-counter antifungal use can delay the correct diagnosis and can contribute to irritant symptoms.

Trap two: Pap tests.

Pap tests are cervical cytology screening tests. They are not reliable vaginitis tests. A Pap report that suggests Candida, bacterial vaginosis, or trichomonas should not be treated as a definitive vaginitis diagnosis.

If a Pap suggests trichomonas, confirmatory diagnostic testing should be performed because of the clinical and partner implications. If a Pap incidentally suggests Candida or bacterial vaginosis in an asymptomatic patient, treatment is not required just because cytology mentioned it.

Trap three: asymptomatic colonization.

Candida can be present without disease. Bacterial vaginosis can be asymptomatic. The resident has to diagnose the patient, not only the test result.

Trap four: noninfectious mimics.

Vulvar dermatoses, contact irritants, genitourinary syndrome of menopause, and desquamative inflammatory vaginitis can all look like infection to the patient and sometimes to the clinician. Products matter: soaps, perfumes, wipes, spermicides, lubricants, povidone iodine, and topical medications can all irritate vulvar or vaginal tissues.

When the story is "yeast keeps coming back," ask whether yeast was ever objectively shown. If it was not, the problem may not be recurrent yeast. It may be recurrent treatment of the wrong disease.

Trap five: mixed etiologies.

More than one process can be present. A patient can have bacterial vaginosis and candidiasis. A patient can have infection plus irritant dermatitis. If one positive test does not explain the whole exam, keep thinking.

Return to the four-layer model

Layer one is location.

Vulvar itch, fissures, plaques, and external burning point you toward vulvar disease, candidiasis, irritants, or dermatoses. Vaginal odor and discharge point you toward the vaginal ecosystem. Cervical findings, postcoital bleeding, pelvic pain, or sexually transmitted infection risk may require a broader cervicitis or pelvic inflammatory disease evaluation.

Layer two is ecosystem.

Normal reproductive-age pH is usually below four point five. Elevated pH means the acidic lactobacillus-dominant environment is disrupted or absent. It does not automatically name the disease.

Layer three is organism.

Bacterial vaginosis is diagnosed by Amsel criteria or Nugent scoring. Trichomonas is best diagnosed by NAAT. Candida requires objective yeast by microscopy, culture, or commercial testing in a symptomatic patient.

Layer four is inflammation.

Bacterial vaginosis is usually dysbiosis without prominent inflammation. Trichomoniasis often has inflammatory discharge. Candidiasis can cause dramatic vulvar inflammation even with normal pH.

If you teach yourself to move through those layers, the bedside evaluation becomes much cleaner.

The final synthesis

Vaginitis diagnosis is not guesswork from discharge color. It is a disciplined evaluation of anatomy, ecology, organism, and tissue response.

Bacterial vaginosis is high pH, dysbiosis, clue cells, whiff, and no routine culture.

Trichomoniasis is a sexually transmitted infection with elevated pH and imperfect wet mount sensitivity; use NAAT when it is plausible.

Candidiasis is symptoms plus objective yeast, often with normal pH; culture when microscopy is negative but symptoms persist, or when recurrent or resistant disease makes species important.

And the last resident reflex is the simplest one.

Before treating discharge, test the environment.